

Design Note: News Recommendation Pipeline

CS4.406: Information Retrieval & Extraction — Assignment 1

Lexical & Semantic Retrieval Baselines on MIND and EB-NeRD

1. System Architecture & Key Design Choices

We built a reproducible, scalable news recommendation retrieval framework covering both **MIND** and **EB-NeRD** (demo + small). The pipeline consists of five modular stages: (1) Unified Data Pipeline & Feature Store, (2) Lexical Candidate Generation (BM25), (3) Semantic Candidate Generation (FAISS/BERT), (4) Offline Evaluation Harness, and (5) Codabench Prediction Generator. A single `make data` rebuilds all datasets from raw files.

Key Architectural Choices:

- Polars + PyArrow Streaming:** Prevents OOM on EB-NeRD’s 13.5M impressions (1.5 GB+) via `scan_parquet` lazy frames and `read_row_group` chunked batching.
- True Inverted Index BM25:** Custom posting-list implementation bypasses `rank_bm25`’s $O(N)$ scans, achieving $\sim 1,000\times$ speedup.
- FAISS IndexFlatIP + BLAS:** Vectorized dot-product ranking (`cand_matrix @ query_vec`) for semantic candidate scoring.
- Temporal Splitting:** Strict time-based splitting ensures no future-click leakage. Unit tests assert `val_max ≤ test_min`.

2. Alternatives Considered & Rationale

Dimension	Alternative	Chosen	Rationale
Data Processing	Pandas	Polars + PyArrow	Pandas OOMs on 13.5M rows; Polars <8 GB
Lexical Engine	<code>rank_bm25</code>	Inverted Index	<code>rank_bm25</code> scans 120k docs/term; index hits only
Candidate Ranking	<code>np.argsort</code>	Timsort	NumPy overhead on tiny lists ($N \approx 30$); Timsort $10\times$
Compression	Deflate L6	LZMA/BZIP2	Deflate >90 MB for 2.37M lines; LZMA <50 MB

3. Experimental Observations

3.1 Retrieval Recall@K

Dataset	Model	R@50	R@100	R@200
EB-NeRD	BM25	0.0200	0.0270	0.0410
EB-NeRD	Semantic	0.0140	0.0230	0.0400
MIND	BM25	0.0088	0.0141	0.0208
MIND	Semantic	0.0125	0.0195	0.0285

3.2 Slice-Level Comparison (Cold-Start ≤ 5 clicks vs. Warm > 5)

- BM25 on cold-start users:** Recall drops significantly since the query is formed from click history—fewer clicks $\rightarrow$ weaker query signal.
- Semantic on cold-start:** Also degrades but less steeply, because mean-pooled embeddings from even 2–3 articles provide a reasonable topical representation.
- Warm users:** Both methods benefit from richer history, but semantic retrieval shows larger absolute gains on MIND (paraphrase-heavy English headlines) while BM25 edges out on EB-NeRD (keyword-specific Danish headlines).

3.3 Key Insights

- Lexical wins on Danish (EB-NeRD):** Headlines contain specific proper nouns (names, places) that exact matching captures well.
- Semantic wins on English (MIND):** Headlines rely on paraphrasing/thematic similarity (“economy” $\leftrightarrow$ “stock market”).
- Cold-start deficit:** Users with ≤ 5 clicks show $AUC \approx 0.52$ vs. warm users $AUC \approx 0.64$.

4. Anti-Gaming Ablation (Q9)

We report metrics **with and without** user click history to verify that our pipeline does not rely on features unavailable at serving time. With an empty history, both retrievers produce uniform/zero scores, reducing to random ranking:

Metric	With History	Without History	Δ
AUC	$\sim 0.54\text{--}0.60$	~ 0.50	$+0.04\text{--}0.10$
MRR	$\sim 0.31\text{--}0.35$	$\sim 0.25\text{--}0.30$	$+0.04\text{--}0.06$
nDCG@10	$\sim 0.35\text{--}0.44$	$\sim 0.30\text{--}0.38$	$+0.04\text{--}0.06$

The positive Δ confirms that our metrics genuinely reflect the retriever’s use of user history, and no inflated signal comes from serving-time-unavailable features. Full results are saved in `outputs/eval_harness_results.json` (generated via `make evaluate-ablation`).

5. Breakdown at $10\times$ Scale

At 135M impressions and 1.2M articles:

- Embedding Bottleneck:** $1.2\text{M} \times 768\text{-dim float32} = \sim 3.7\text{ GB RAM}$. FAISS `IndexFlatIP` slows linearly. *Fix:* FAISS `IndexIVFPQ` ($8\times$ memory reduction).
- BM25 Dict Overhead:** Python `dict` posting lists exceed 8 GB. *Fix:* Pyserini/Lucene disk-backed engine.
- Sequential I/O:** Writing 135M lines takes $>2\text{ h}$. *Fix:* 16-process multiprocessing chunking + binary concat.

6. Codabench Leaderboard Submissions

ID # ▾	File name	Date	Status	Score	Actions
898624	mind_semantic_submission.zip	2026-08-24 00:47	Finished	0.6129	✓
898369	mind_bm25_submission.zip	2026-08-23 20:40	Finished	0.5632	

Figure 1: MIND Codabench leaderboard submission scores.

ID # ▾	File name	Date	Status	Score	Detailed Results	Actions
900048	ebnerd_semantic_submission.zip	2026-08-24 21:31	Submitted			⊖
898694	ebnerd_bm25_submission.zip	2026-08-24 02:16	Finished	0.5097		✓

Figure 2: EB-NeRD (RecSys 2024 Challenge) Codabench leaderboard submission scores.